
Assignment

Implementing user program & system call

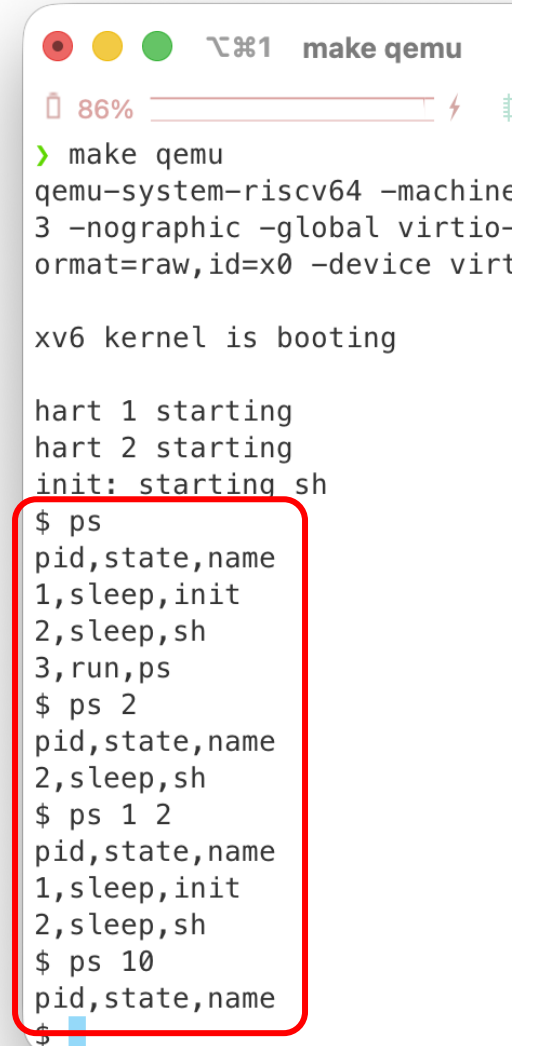
Due date
2026. 04. 01. 23:59

Overview

- **Goal**
 - Implement the ps **user-level program** and a **system call** to display the status of currently active processes in xv6.
- **Submission Guidelines**
 - Submit the modified **xv6 source code** and **report** of this assignment.
 - Push all your source code and report to the following GitHub Classroom repository:
 - <https://classroom.github.com/a/ISY9EvQr>
- **Deadline: April 1, 2026 (Wed) 23:59**
 - Late submissions are allowed, but your final score will be reduced by **5%** for each **3-hour window** following the due date.
 - Grading is based on your **final commit**. Be extremely cautious about committing after the deadline, as it will trigger the late penalty.
- There will be a **short quiz** during the first class following the assignment deadline.

Specifications

- Specifications of the ps user-level program
 - Print processes information based on arguments:
 - **No arguments:** Displays the information of **all active processes**.
 - **With arguments:** Displays the information of the **specific process(es)** that matches the given **PID(s)**.
 - **Printing format:**
 - Print the header **pid,state,name** on the first line.
 - State strings: **unused** / **used** / **sleep** / **runble** / **run** / **zombie**
 - Must be comma-separated. Do not add any extra whitespaces, tabs or decorative characters. (e.g., **1,sleep,init**)
 - **Any deviation from this exact format can result zero score.**



```
make qemu
86%
> make qemu
qemu-system-riscv64 -machine
3 -nographic -global virtio-
ormat=raw,id=x0 -device virt

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh

$ ps
pid,state,name
1,sleep,init
2,sleep,sh
3,run,ps
$ ps 2
pid,state,name
2,sleep,sh
$ ps 1 2
pid,state,name
1,sleep,init
2,sleep,sh
$ ps 10
pid,state,name
$
```

Specifications

- **Specifications of ps system call**

- **Role:** Iterates through the kernel's process table to find and print the status of the requested process(es) directly to the console.

- **Signature:**

```
int ps(int pid);
```

- **pid (int):** The PID of the specific process to query. If pid is **0**, it must display the status of all active processes
- **Return value:**
 - Return 0 if the target process(es) exist and are successfully printed.
 - Return 1 if the specified process does not exist.

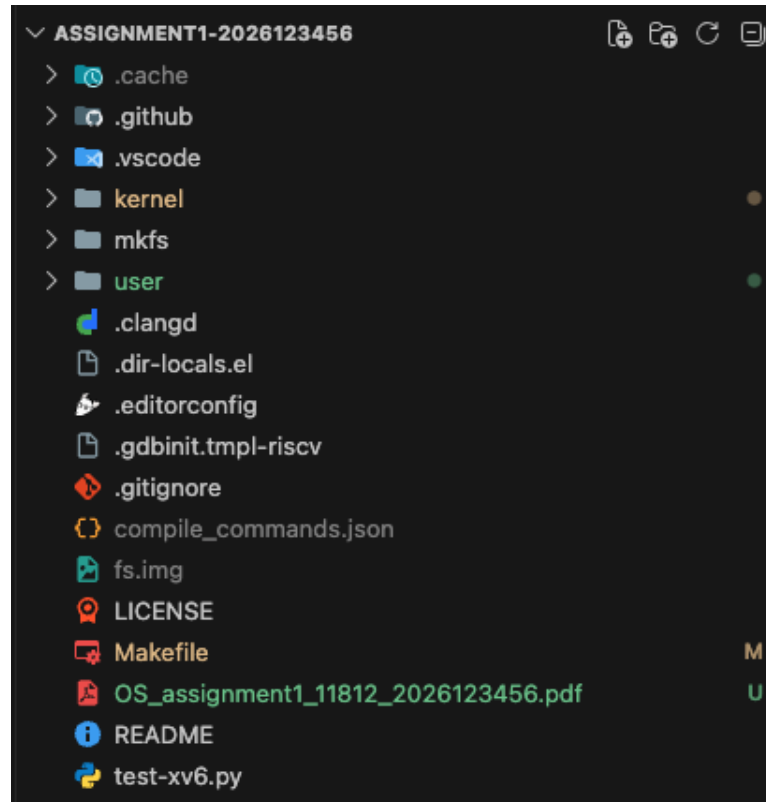
❖ **Hint:** Refer to the **procdump()** function in **kernel/proc.c** for implementation insights.

Report

- **Design**
 - Explain your implementation plan to meet the assignment requirements.
- **Implementation**
 - Detail the specific modifications and additions you made to the given xv6 codebase. Clearly highlight how they differ from the original code and the purpose of these changes.
- **Results**
 - Document the compilation and execution process. Include screenshots verifying that all requirements are met, alongside explanations of the execution flow.
- **Troubleshooting**
 - If you encountered any issues during the assignment, describe the issues and your resolution process. For unresolved issues, explain what they were and what approaches you attempted.
- **System Call Execution Flow (Only for this assignment)**
 - Describe the step-by-step execution process of the system call, tracing the complete path from the user-level program invocation down to the kernel-level execution and back.

Report

- The report must be a PDF file strictly named: **OS_assignment1_[class number]_[student ID].pdf**
- You must commit and push your report PDF to your GitHub Classroom repository along with your xv6 source code.
- Make sure that report PDF is in the root directory of your repository, as shown in below.



Q&A

- For assignment related questions, please use the **Discussions tab** of xv6-sandbox repository.
 - <https://github.com/2026-HYU-ELE3021/xv6-riscv-sandbox/discussions>
- **Questions sent by email will not be answered.**
- For non-assignment questions or personal matters, please use an email like before.

Search: is:open category:Questions

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Questions

[assignment1] ps 핵심 로직 구현이 너무 막막합니다. **예시**
ESeokhwan asked 3 minutes ago in Questions · Answered

Most helpful Last 30 days

- Students are highly **encouraged to answer each other's questions**! Active participation is greatly appreciated and helpful for everyone

Test Case Contribution

- You can share your custom test cases on the **Discussions tab** of xv6-sandbox repository too.
 - <https://github.com/2026-HYU-ELE3021/xv6-riscv-sandbox/discussions>
- A structured template is already provided, so we highly encourage your active contributions.
- Well-crafted test cases might actually be adopted for the official grading process!

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Q is:open category:test-cases

Categories

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- test-cases**

test-cases

1 **[lab03] getppid usertests**
wnsah814 started 1 hour ago in test-cases 0

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test-cases

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Add a title

[과제 이름{assignment1, proj1, proj2, ...}] <테스트할 기능이나 버그를 요약해주세요>

1. 테스트 목적 (Objective)

이 테스트 케이스가 무엇을 검증하기 위한 것인지 간략하게 설명해 주세요.

예) 이 테스트는 파이프가 꽂았을 때 부모와 자식 간의 데드락이 발생하는지 검증합니다.

2. 테스트 코드 (Test Code)

user/usertest.c 에 추가하거나 독립적인 유저 프로그램으로 실행할 C 코드를 작성해 주세요.

```
```c
int
my_custom_test(char *s)
{
```

**3. 실행 방법 (How to Run)**

다른 학우들이 이 테스트를 어떻게 실행해 볼 수 있는지 명령어를 적어주세요.

1. user/usertest.c 에 테스트 코드 함수를 붙여넣고...
2. make qemu 후 \$ usertests custom\_test...